

# Core Components Assessment

GCSE Computer Science · Python Lab · Archbishop McGrath Catholic High School

Xander n · Class not entered · 10CS · 2026-09-24



**54 / 56 marks · 96.4%** · 2 not yet achieved · 0 unmarked  
 Mastery · Teacher-reviewed classroom scheme

Developing: 25% (14 marks) · Secure: 60% (34 marks) · Mastery: 80% (45 marks). These are classroom checkpoints, not GCSE grades. Unmarked criteria have not been treated as incorrect.

## Assessment breakdown

| Task                    | Achieved | Not yet | Unmarked |
|-------------------------|----------|---------|----------|
| A · Knowledge & tracing | 7 / 9    | 2       | 0        |
| B · Debug lab           | 6 / 6    | 0       | 0        |
| C1 · Merchandise order  | 12 / 12  | 0       | 0        |
| C2 · Priority delivery  | 8 / 8    | 0       | 0        |
| C3 · Booking repair     | 9 / 9    | 0       | 0        |
| C4 · Tournament budget  | 12 / 12  | 0       | 0        |

## A message from your teacher

Xander, this is an excellent result and shows a very strong understanding of the programming skills covered.

You have worked accurately and confidently throughout the assessment and demonstrated secure knowledge across almost all areas. Keep up this high standard and continue to challenge yourself with more complex programming tasks.

## WWW · What went well

Calculations and tracing

You selected appropriate operators and linked stored values to calculate results.

Evidence: C1.5, C1.6.

Inputs · input()

You collected and stored the values needed by the program.

Evidence: C1.1, C1.2.

Type casting · int() and float()

You selected an appropriate numerical conversion in the recorded work.

Evidence: C1.3, C1.4.

Naming conventions and comments

You used meaningful identifiers or useful comments to make your program easier to follow.

Evidence: C1.11, C1.12.

## EBI · Even better if

Outputs · print()

Use print() to display each required result. Distinguish literal text in quotes from a variable's stored value.

Evidence: A5.1.

Variables

Trace each variable from its assignment to its next use. Check that you use the correct stored value in each calculation and output.

Evidence: A5.1.

Python knowledge and error types

Use precise Python vocabulary. Distinguish the language-rule fault in a syntax error from a logic error that runs but gives the wrong result.  
Evidence: A8.2.

## Individual mark checklist

Each criterion shows its maximum mark. "Not yet" is a missed mark; "Unmarked" awaits a decision. This is a proposed teacher scheme based on the workbook's questions and requirements, not an official exam-board scheme.

### A1 · Question 01 · 1 / 1

| Criterion | What was required                 | Decision |
|-----------|-----------------------------------|----------|
| A1.1      | Give the correct answer: Boolean. | 1 / 1    |

### A2 · Question 02 · 1 / 1

| Criterion | What was required                    | Decision |
|-----------|--------------------------------------|----------|
| A2.1      | Give the correct answer: firstScore. | 1 / 1    |

### A3 · Question 03 · 1 / 1

| Criterion | What was required                                   | Decision |
|-----------|-----------------------------------------------------|----------|
| A3.1      | Give the correct answer: age = int(input("Age: ")). | 1 / 1    |

### A4 · Question 04 · 1 / 1

| Criterion | What was required           | Decision |
|-----------|-----------------------------|----------|
| A4.1      | Give the correct answer: 3. | 1 / 1    |

### A5 · Question 05 · 0 / 1

| Criterion | What was required                           | Decision |
|-----------|---------------------------------------------|----------|
| A5.1      | Give the correct answer: print(workerName). | 0 / 1    |

### A6 · Question 06 · 1 / 1

| Criterion | What was required           | Decision |
|-----------|-----------------------------|----------|
| A6.1      | Give the correct answer: 4. | 1 / 1    |

### A7 · Question 07 · 1 / 1

| Criterion | What was required                        | Decision |
|-----------|------------------------------------------|----------|
| A7.1      | Give the correct answer: Inspector: Sam. | 1 / 1    |

### A8 · Question 08 · 1 / 2

| Criterion | What was required                                                                                       | Decision |
|-----------|---------------------------------------------------------------------------------------------------------|----------|
| A8.1      | Explain that a syntax error breaks Python's language rules and prevents the affected code from running. | 1 / 1    |
| A8.2      | Explain that a logic error allows a program to run but produces the wrong result.                       | 0 / 1    |

**Teacher note:** Expand on this more. What is incorrect code? Give an example.

### B1 · Debug 01 · Merchandise order · 3 / 3

| Criterion | What was required                                                       | Decision |
|-----------|-------------------------------------------------------------------------|----------|
| B1.1      | Convert the quantity input to an integer before using it in arithmetic. | 1 / 1    |
| B1.2      | Use quantity × unitPrice to calculate the total.                        | 1 / 1    |

| Criterion | What was required                                                   | Decision |
|-----------|---------------------------------------------------------------------|----------|
| B1.3      | Use an f-string to combine the output text with the total variable. | 1 / 1    |

**B2 · Debug 02 · Time converter · 3 / 3**

| Criterion | What was required                                                      | Decision |
|-----------|------------------------------------------------------------------------|----------|
| B2.1      | Use integer division by 60 to calculate complete hours.                | 1 / 1    |
| B2.2      | Use remainder after division by 60 to calculate the remaining minutes. | 1 / 1    |
| B2.3      | Produce the correct labelled result: 2 hours and 22 minutes.           | 1 / 1    |

**C1 · Practical 01 · Gaming merchandise order · 12 / 12**

| Criterion | What was required                                                   | Decision |
|-----------|---------------------------------------------------------------------|----------|
| C1.1      | Collect and store the customer's name.                              | 1 / 1    |
| C1.2      | Collect and store the product name.                                 | 1 / 1    |
| C1.3      | Collect quantity and convert it to an integer.                      | 1 / 1    |
| C1.4      | Collect unit price and convert it to a float.                       | 1 / 1    |
| C1.5      | Calculate subtotal as quantity × unit price.                        | 1 / 1    |
| C1.6      | Use a fixed delivery charge of £2.50.                               | 1 / 1    |
| C1.7      | Calculate total as subtotal + delivery.                             | 1 / 1    |
| C1.8      | Identify the customer, product and quantity in the order summary.   | 1 / 1    |
| C1.9      | Display labelled subtotal, delivery and total values.               | 1 / 1    |
| C1.10     | Format every monetary amount to two decimal places.                 | 1 / 1    |
| C1.11     | Add useful section comments.                                        | 1 / 1    |
| C1.12     | Use meaningful variable names that make the program easy to follow. | 1 / 1    |

**C2 · Practical 02 · Priority delivery upgrade · 8 / 8**

| Criterion | What was required                                                            | Decision |
|-----------|------------------------------------------------------------------------------|----------|
| C2.1      | Retain the existing item, quantity and unit-price inputs with correct types. | 1 / 1    |
| C2.2      | Retain the correct subtotal calculation.                                     | 1 / 1    |
| C2.3      | Store a fixed priority-delivery charge of £6.00.                             | 1 / 1    |
| C2.4      | Calculate the updated total as subtotal + priority delivery.                 | 1 / 1    |
| C2.5      | Display a PRIORITY ORDER heading.                                            | 1 / 1    |
| C2.6      | Show the item and a clear subtotal, delivery and total breakdown.            | 1 / 1    |
| C2.7      | Display all monetary values to two decimal places.                           | 1 / 1    |
| C2.8      | Add a useful comment explaining the new section.                             | 1 / 1    |

**C3 · Practical 03 · Gaming lounge booking repair · 9 / 9**

| Criterion | What was required                          | Decision |
|-----------|--------------------------------------------|----------|
| C3.1      | Convert the number of teams to an integer. | 1 / 1    |
| C3.2      | Convert players per team to an integer.    | 1 / 1    |
| C3.3      | Convert price per player to a float.       | 1 / 1    |

| Criterion | What was required                                                       | Decision |
|-----------|-------------------------------------------------------------------------|----------|
| C3.4      | Calculate seats as teams × players per team.                            | 1 / 1    |
| C3.5      | Calculate total price as seats × price per player.                      | 1 / 1    |
| C3.6      | Repair the seat output so it displays without a text/number type error. | 1 / 1    |
| C3.7      | Repair the total-price output so it displays without a type error.      | 1 / 1    |
| C3.8      | Display the total price as money to two decimal places.                 | 1 / 1    |
| C3.9      | Add useful comments explaining the repaired program.                    | 1 / 1    |

#### C4 · Practical 04 · Sports tournament budget · 12 / 12

| Criterion | What was required                                                     | Decision |
|-----------|-----------------------------------------------------------------------|----------|
| C4.1      | Collect and store the tournament name.                                | 1 / 1    |
| C4.2      | Collect team count and players per team as appropriate whole numbers. | 1 / 1    |
| C4.3      | Collect entry price per player as a decimal-capable number.           | 1 / 1    |
| C4.4      | Collect venue hire and officials cost as decimal-capable numbers.     | 1 / 1    |
| C4.5      | Calculate total players as teams × players per team.                  | 1 / 1    |
| C4.6      | Calculate entry income as total players × entry price.                | 1 / 1    |
| C4.7      | Calculate total costs as venue hire + officials cost.                 | 1 / 1    |
| C4.8      | Calculate the expected balance as income – total costs.               | 1 / 1    |
| C4.9      | Display a complete, clearly labelled budget summary.                  | 1 / 1    |
| C4.10     | Format the monetary values to two decimal places.                     | 1 / 1    |
| C4.11     | Use purposeful comments to organise the budget program.               | 1 / 1    |
| C4.12     | Use meaningful identifiers throughout the budget program.             | 1 / 1    |

## Progress rubric

Developing means some requirements achieved; Secure means the key concepts achieved; Mastery means all checklist requirements for that skill achieved. The green tick shows the selected level. Skill levels are separate from the overall checkpoint badge.

| Skill                                              | Developing                                                      | Secure                                                                                                       | Mastery                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs · input()</b><br>Mastery                 | I can collect a value from the user and store it in a variable. | I can use clear prompts to collect and store the main information a program needs.                           | ✓ I can collect and store all the information needed for each program, including names, quantities, prices and costs.                           |
| <b>Outputs · print()</b><br>Mastery                | I can display text and the value stored in a variable.          | I can display the main results with clear labels so the user can understand them.                            | ✓ I can produce complete, clearly labelled summaries that show every requested detail and result, with money formatted where required.          |
| <b>Variables</b><br>Mastery                        | I can create variables and store values in them.                | I can store inputs and calculation results in variables, then reuse those values in calculations and output. | ✓ I can assign, update and reuse values accurately throughout my programs, and correctly trace how variable values change.                      |
| <b>Naming conventions and comments</b><br>Mastery  | I can use valid variable names and add a simple comment.        | I can choose meaningful names and use comments to identify the main sections of my code.                     | ✓ I can make my programs easy to follow, using meaningful names throughout and useful comments for the sections, changes and repairs requested. |
| <b>Type casting · int() and float()</b><br>Mastery | I can convert an input into a number using int() or float().    | I can use int() for whole-number counts and float() for prices so the values can be used in calculations.    | ✓ I can apply suitable conversions to every numerical input, handling whole-number quantities and decimal prices and costs correctly.           |

| Skill                                              | Developing                                                                           | Secure                                                                                                                                     | Mastery                                                                                                                                                                |
|----------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Calculations and tracing</b><br>Mastery         | I can carry out simple arithmetic and follow a short sequence of calculations.       | I can choose the correct operators to calculate quantities, subtotals, totals and remainders.                                              | ✓ I can build and trace the required calculations accurately from start to finish, including the time conversion and the tournament's income, costs and final balance. |
| <b>F-strings and formatting</b><br>Mastery         | I can recognise an f-string and use one to include a variable in a simple message.   | I can combine text and variable values correctly using the f prefix and braces {}.                                                         | ✓ I can correctly predict and produce f-string output, and display monetary values to two decimal places wherever the task requires it.                                |
| <b>Python knowledge and error types</b><br>Mastery | I can recognise basic Python terms, including Boolean values and programming errors. | I can identify Boolean values and distinguish syntax errors from logic errors, although my description of their effects may be incomplete. | ✓ I can correctly identify Boolean values and describe how syntax errors break Python's rules and prevent code running, while logic errors produce incorrect results.  |
| <b>Debugging and repair</b><br>Mastery             | I can identify and correct a simple mistake in supplied code.                        | I can repair common mistakes in input types, calculation operators and output statements.                                                  | ✓ I can fully repair the supplied programs so their inputs, calculations and outputs work together correctly, while retaining the code that already works.             |

## DIRT · Improve your work

[ ] **Your improvement task:** Extend your knowledge using the unit 1 Challenge Lab. This will introduce string manipulations and ask you to apply it in python code.

**What did you change?**

**You can find my improved work/DIRT task at**

DIRT completion does not change the original assessment mark. Keep pupil information private.